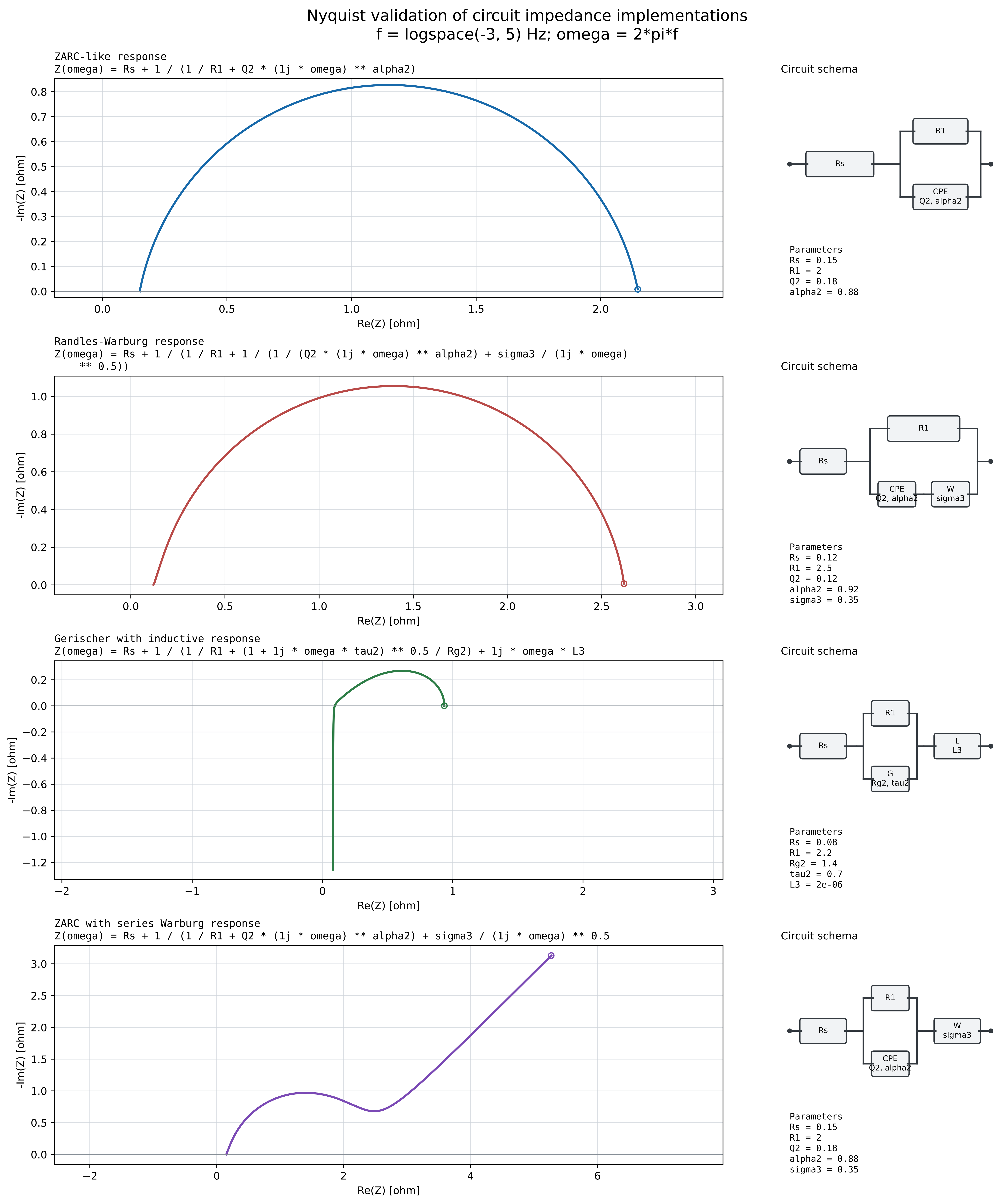
Nyquist validation of circuit impedance implementations
f = logspace(-3, 5) Hz; omega = 2*pi*f
ZARC-like response
Z(omega) = Rs + 1 / (1 / R1 + Q2 * (1j * omega) ** alpha2)
-Im(Z) [ohm]
Re(Z) [ohm]
Circuit schema
Rs
R1
CPE
Q2, alpha2
Parameters
Rs = 0.15
R1 = 2
Q2 = 0.18
alpha2 = 0.88
Randles-Warburg response
Z(omega) = Rs + 1 / (1 / R1 + 1 / (1 / (Q2 * (1j * omega) ** alpha2) + sigma3 / (1j * omega) ** 0.5))
-Im(Z) [ohm]
Re(Z) [ohm]
Circuit schema
Rs
R1
CPE
Q2, alpha2
W
sigma3
Parameters
Rs = 0.12
R1 = 2.5
Q2 = 0.12
alpha2 = 0.92
sigma3 = 0.35
Gerischer with inductive response
Z(omega) = Rs + 1 / (1 / R1 + (1 + 1j * omega * tau2) ** 0.5 / Rg2) + 1j * omega * L3
-Im(Z) [ohm]
Re(Z) [ohm]
Circuit schema
Rs
R1
G
Rg2, tau2
L
L3
Parameters
Rs = 0.08
R1 = 2.2
Rg2 = 1.4
tau2 = 0.7
L3 = 2e-06
ZARC with series Warburg response
Z(omega) = Rs + 1 / (1 / R1 + Q2 * (1j * omega) ** alpha2) + sigma3 / (1j * omega) ** 0.5
-Im(Z) [ohm]
Re(Z) [ohm]
Circuit schema
Rs
R1
CPE
Q2, alpha2
W
sigma3
Parameters
Rs = 0.15
R1 = 2
Q2 = 0.18
alpha2 = 0.88
sigma3 = 0.35